ECO 344 - Applied Econometrics I
Chapter 6

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Outline

1. Data Scaling
2. More Functional Forms
3. More on Goodness of Fit & Selection of Regressors
4. Prediction & Residual Analysis

Data Scaling

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- The good news is that nothing of substance changes. Scaling is only a change of units.

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  - Because $t = \hat{\beta}_j / se(\hat{\beta}_j)$ and both the numerator and the denominator are multiplied by c.

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- Because $F = \dfrac{R^2/k}{(1-R^2)/(n-k-1)}$ and R^2 has not changed.

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  - The intercept and all the other coefficients and standard errors stay the same.

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- The F-statistic does not change. Why?
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- In short, scaling x rescales its own slope and standard error. Scaling y rescales everything in units of y.

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- This is why the units of a logged variable never matter for the slopes.

Example

- Let's go back to the example that we covered in one of the labs: "The Effect of Smoking on Birth Weight". We are interested in the following model:

$$bwght = \beta_0 + \beta_1 cigs + \beta_2 faminc + \mu$$

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- Similarly, for occasional smokers the number of cigarettes makes sense, but for chain smokers the number of packs makes more sense. (1 pack = 20 cigarettes)

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- The next two slides show the regression with different units of birth weight and cigarettes.

Example

TABLE 6.1 Effects of Data Scaling

Dependent Variable	(1) *bwght*	(2) *bwghtlbs*	(3) *bwght*
Independent Variables			
cigs	–.4634 (.0916)	–.0289 (.0057)	—
packs	—	—	–9.268 (1.832)
faminc	.0927 (.0292)	.0058 (.0018)	.0927 (.0292)
intercept	116.974 (1.049)	7.3109 (.0656)	116.974 (1.049)
Observations	1,388	1,388	1,388
R-Squared	.0298	.0298	.0298
SSR	557,485.51	2,177.6778	557,485.51
SER	20.063	1.2539	20.063

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Dependent	*bwght*	*bwghtlbs*	*bwght*
Independent	*cigs*	*cigs*	*packs*
cigs	-0.4634 (0.0916)	$-0.4634/16 = -0.0289$ $(0.0916/16 = 0.0057)$	-
packs	-	-	$-0.4634 \times 20 = -9.268$ $(0.0916 \times 20 = 1.832)$
faminc	0.0927 (0.0292)	$0.0927/16 = 0.0058$ $(0.0292/16 = 0.0018)$	0.0927 (0.0292)
intercept	116.974 (1.049)	$116.974/16 = 7.3109$ $(1.049/16 = 0.0656)$	116.974 (1.049)
SSR	557,485.51	$557{,}485.51/16^2 = 2{,}177.68$	557,485.51
SER	20.063	$20.063/16 = 1.2539$	20.063
R^2	0.0298	0.0298	0.0298

Standard errors in parentheses. $n = 1{,}388$ in all three columns.

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- The t-statistic of *cigs* is the same in all three columns.

$$\frac{-0.4634}{0.0916} \approx \frac{-0.0289}{0.0057} \approx \frac{-9.268}{1.832} \approx -5.06$$

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- Conclusion: the choice of units is a presentation choice. It never changes what the data say.

Standardized Coefficients

- We read about standardization in Chapter 2 and Chapter 3.

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- However, we can use the standardized SAT score. This tells us the effect of a 1 standard deviation change in SAT score on the dependent variable.
- It also lets us compare which independent variable matters more, since all of them are now measured in standard deviations.

Standardized Coefficients

- The regression with standardized variables looks like

$$\frac{y-\bar{y}}{\sigma_y} = b_0 + b_1\frac{x_1-\bar{x}_1}{\sigma_{x_1}} + b_2\frac{x_2-\bar{x}_2}{\sigma_{x_2}} + \cdots + b_k\frac{x_k-\bar{x}_k}{\sigma_{x_k}} + \text{error}$$

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- Interpretation: a one standard deviation increase in x_j leads to a b_j standard deviation change in y, holding the other variables constant.

Standardized Coefficients

- How can we get the standardized coefficients without running the regression on standardized variables?

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- We start with the model

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- Re-arranging:

$$y - \bar{y} = \beta_1 (x_1 - \bar{x}_1) + \beta_2 (x_2 - \bar{x}_2) + \cdots + \beta_k (x_k - \bar{x}_k) + \mu$$

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- Divide both sides by σ_y.

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- Multiply and divide each term on the R.H.S. by its own σ_{x_j} for $j = 1, 2, \ldots, k$.

$$\frac{y - \bar{y}}{\sigma_y} = \frac{\sigma_{x_1}}{\sigma_y}\beta_1\frac{(x_1 - \bar{x}_1)}{\sigma_{x_1}} + \cdots + \frac{\sigma_{x_k}}{\sigma_y}\beta_k\frac{(x_k - \bar{x}_k)}{\sigma_{x_k}} + \frac{\mu}{\sigma_y}$$

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More Functional Forms

Logarithmic Functional Forms

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- There are problems with using logarithms that we should be aware of, but first let's look at the benefits.

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- Strictly positive variables with large values and outliers are generally transformed to logs (wage, population, sales, firm size, house price and so on).
- Rule of thumb: do not log variables measured in years (education, experience, age) or variables already in percentages or proportions (unemployment rate, interest rate).

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3. A log transformation is not a linear rescaling. Unlike data scaling, the coefficients, standard errors, sums of squares, R^2, t-tests and F-tests all change and we cannot recover them without re-running the regression.

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- If y is in logs and x is a percentage, then a 1 percentage point increase in x changes y by $100\beta_1$ percent.

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Model with Quadratics

- Let we have the following estimates after running the model

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$$\widehat{wage} = -3.96 + 0.6\, educ + 0.27\, exper - 0.004\, exper^2$$

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  - For an individual with 20 years of experience the marginal effect is 0.11 USD.
  - At what level of experience is the effect zero? At $exper = \frac{0.27}{0.008} = 33.75$ years. Beyond that, the effect is negative.

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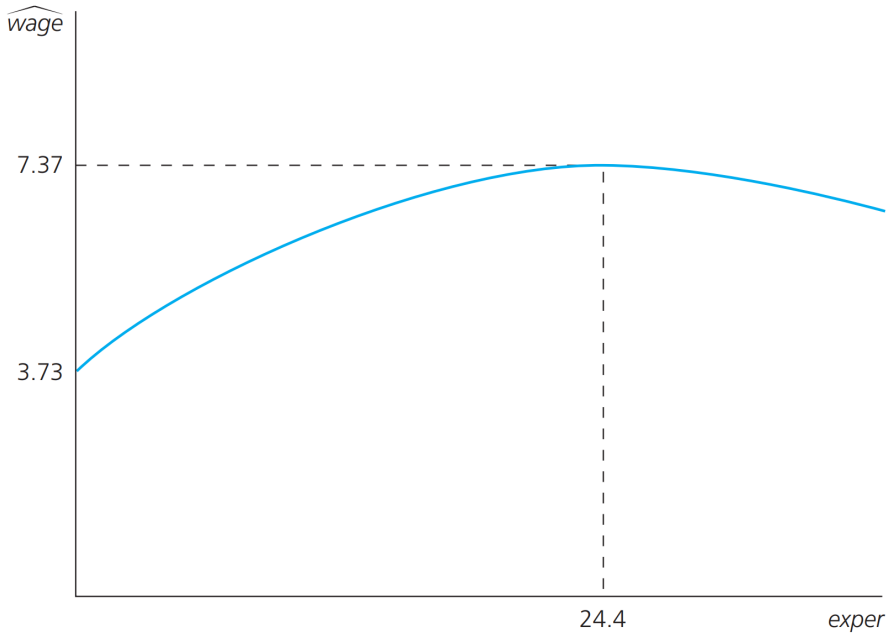
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- Up to 24.4 years more experience raises the predicted wage. After that, more experience lowers it.
- Always check how many observations lie beyond the turning point. If almost none do, the downward part of the curve is not something the data can speak about.

Model with Quadratics

FIGURE 6.1 Quadratic relationship between $\widehat{wage}$ and *exper*.



Model with Quadratics

Average Partial Effect

- As the effect of x_1 on y depends on x_1 itself, what is the simplest way to summarize the effect of x_1 on y?

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- So on average, one more year of experience raises the wage by about 0.134 USD.

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- Always think about the interpretation of the coefficients when using interaction terms, as it depends on the context.

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- For practice, go over Example 6.3 in the book.

More on Goodness of Fit & Selection of Regressors

Adjusted R-Squared

- We have studied that R^2 never falls when we add independent variables, regardless of their relevance.

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- There is now a penalty for adding an independent variable. $\bar{R}^2$ rises only if SSR falls by enough to offset the lost degree of freedom.
- In fact, $\bar{R}^2$ rises when a variable is added if and only if the absolute value of its t-statistic is greater than 1.

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- $\bar{R}^2$ can increase or decrease when an independent variable is added, depending on how much the variable reduces SSR.
- $\bar{R}^2$ is always smaller than R^2 when $k \geq 1$, and it can be negative. A negative value indicates a very poor fit relative to the degrees of freedom.

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- Because the F-test works by imposing restrictions on a model. Here neither model is a restricted version of the other.
- We can compare the adjusted R^2 of the two models instead. The model with the higher $\bar{R}^2$ fits better after the penalty. This works only if both models have the same dependent variable. We cannot compare $\bar{R}^2$ of a model for y with a model for $\log(y)$.

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  - Example: if we want the effect of beer taxes on traffic fatalities, we should not control for beer consumption. The tax works through consumption.

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- A smaller error variance gives us a lower sampling variance and thus lower standard errors for the estimators, leading to more precise estimates.
- This is especially useful in experiments where the treatment is randomly assigned.

Prediction & Residual Analysis

Confidence Interval of $\hat{y}$

- As we know, we can find a confidence interval by

$$\hat{c} - t_{\frac{\alpha}{2}} \times se(\hat{c}) \leq c \leq \hat{c} + t_{\frac{\alpha}{2}} \times se(\hat{c})$$

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- So we need a trick.

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- Take a point $(c_1, c_2, \ldots, c_k)$. In the population, the expected value of y at that point is

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$$y = \theta_0 + \beta_1(x_1 - c_1) + \beta_2(x_2 - c_2) + \cdots + \beta_k(x_k - c_k) + \mu$$

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- So the intercept in this regression is θ_0, and its standard error is $se(\hat{\theta}_0)$.

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6. This is not a general formula. You have to repeat it for every point at which you want a confidence interval.

Confidence Interval of $\hat{y}$ - Example

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  4. The 95% confidence interval is

$$2.70 - 1.96 \times 0.020 \leq E[colGPA] \leq 2.70 + 1.96 \times 0.020 \Rightarrow [2.66,\ 2.74]$$

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- This gives a **prediction interval**, which is wider than the confidence interval for the average.

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- This is because $E[\hat{\beta}_j] = \beta_j$ and $E[\mu^0] = 0$.

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- What is the variance of the prediction error?

$$\hat{e}^0 = (\beta_0 - \hat{\beta}_0) + (\beta_1 - \hat{\beta}_1)x_1^0 + \cdots + (\beta_k - \hat{\beta}_k)x_k^0 + \mu^0$$

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- The error μ^0 is not part of our sample, so it is uncorrelated with the $\hat{\beta}_j$. Therefore

$$Var(\hat{e}^0) = Var(\hat{y}^0) + Var(\mu^0) = Var(\hat{y}^0) + \sigma^2$$

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$$\hat{e}^0 = (\beta_0 - \hat{\beta}_0) + (\beta_1 - \hat{\beta}_1)x_1^0 + \cdots + (\beta_k - \hat{\beta}_k)x_k^0 + \mu^0$$

- The error μ^0 is not part of our sample, so it is uncorrelated with the $\hat{\beta}_j$. Therefore

$$Var(\hat{e}^0) = Var(\hat{y}^0) + Var(\mu^0) = Var(\hat{y}^0) + \sigma^2$$

- The standard error of the prediction error is

$$se(\hat{e}^0) = \sqrt{[se(\hat{y}^0)]^2 + \hat{\sigma}^2}$$

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- $se(\hat{y}^0)$ is found using the same steps as before, and $\hat{\sigma}$ is the standard error of the regression.

Prediction Interval - Example

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- Go over Example 6.6 in the book.

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- When the dependent variable is $\log(y)$, a residual of -0.30 means that the actual value is roughly 30 percent below the predicted value.
- Therefore, residual analysis can generate different insights depending on the context.